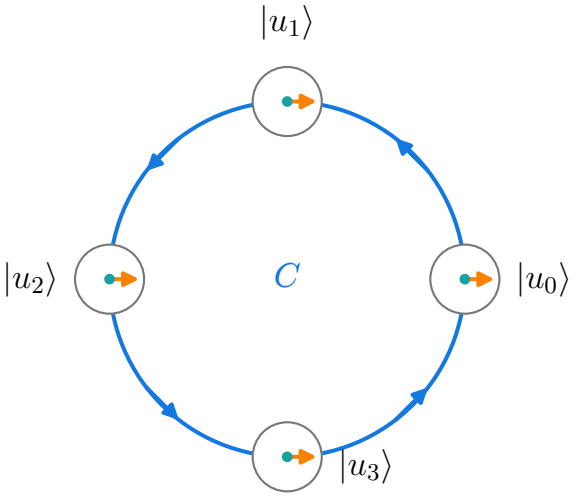
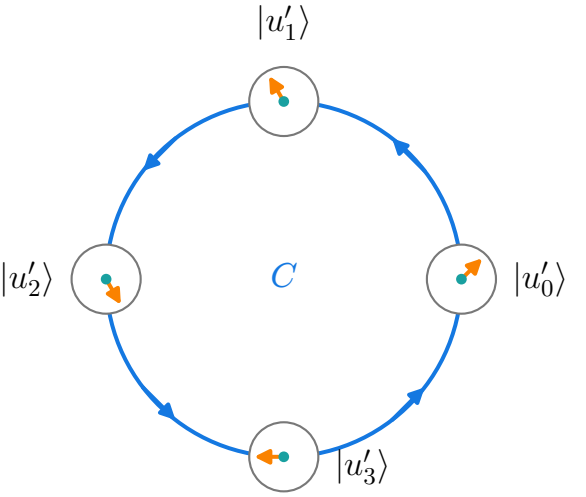
(a) Reference phases



$$Z_C = \prod_j \langle u_j | u_{j+1} \rangle$$

(b) Rephased states



$$Z'_C = Z_C$$